

CLASS 9 · ICSE · SELINA CONCISE MATHEMATICS · DPP SHEET

Chapter 8: Logarithms

Daily practice problems: 10 questions with an answer key

Try every question on your own first. The answer key is on the last page.

1. Write in logarithmic form: $5^3 = 125$ and $10^{-3} = 0.001$.
2. Find $\log_3 243$ and $\log_4 \frac{1}{64}$.
3. Find x if $\log_x 125 = 3$ and if $\log_{16} x = \frac{3}{4}$.
4. Express $\log \frac{x^3}{y^2z}$ in terms of $\log x$, $\log y$, $\log z$.
5. Write as a single logarithm: $\log 6 + 2 \log 5 - \log 15$.
6. Evaluate $\frac{\log 32}{\log 4}$.
7. Given $\log 2 = 0.3010$ and $\log 3 = 0.4771$, find $\log 18$ and $\log 25$.
8. Solve $\log(x + 1) + \log(x - 1) = \log 24$.
9. Solve $\log(3x + 1) = 2$.
10. If $\log(x - y) = \log x - \log y$, express x in terms of y .

Answer Key

1. $\log_5 125 = 3$ and $\log_{10} 0.001 = -3$

2. 5 and -3

3. $x = 5$; $x = 8$

4. $3 \log x - 2 \log y - \log z$

5. $\log 10 = 1$

6. $\frac{5}{2}$

7. 1.2552 and 1.3980

8. $x = 5$

9. $x = 33$

10. $x = \frac{y^2}{y - 1}$